

Can a learned colour prior beat a trivial one?

Part-based RePaint on completed point clouds

Weekly research report 2026-08-19 — 7–19 August 2026

Efe Yenice (researcher) · Claude (AI co-researcher)

method and original notebooks: **Pelin Kılıç** · port, scaling, controls and evaluation: this project

Question of the week: when the geometry is already complete, can a diffusion colour prior colour the missing points better than copying or averaging the visible ones?

TL;DR — the week in three findings

- 1. On real texture the method loses decisively — and we can say exactly why.** The diffusion colour prior scores 61.0 ΔE against 22.7 for NN-copy and 24.5 for part-mean, winning 0 of 34 test models. Four independent controls rule out the mundane explanations (Section 5).
- 2. The reason is structural, not a bug: the prize collapsed.** On real texture the *oracle ceiling* — part-mean given perfect labels, the best *any* part-colour method could achieve — is 18.7, while the best trivial rule already scores 22.7. The entire margin available to this whole class of methods is **4.0 ΔE** (≈ 2 just-noticeable differences), against 33.4 ΔE on synthetic colours: a $8.4\times$ collapse (Figure 1). The method was not merely beaten; it was playing for almost nothing.
- 3. The same code wins on synthetic colours, which is what makes the negative readable.** At 23.5 vs. 38.7/49.1 it wins 99/110 models and stays flat as occlusion grows while the copy rules decay — textbook “a prior helps when evidence disappears”. On real texture that behaviour inverts exactly (Figure 2). Prior *learnability*, not implementation, is the variable.

1 The story so far: why we reached for a generative model

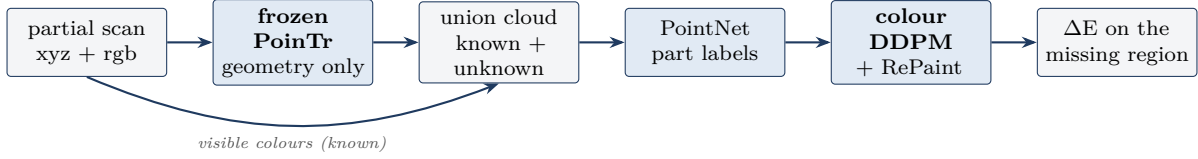
The project asks whether per-point colour helps completion, and under what conditions. Two previous results set up this week. First, the s5 arms: a learned colour head that predicted RGB for the completed cloud never beat *mean-copy* — the trivial rule “paint every missing point the average colour of the visible ones”. Second, the s6 data-scale curve: at $N=150$ models/category the learned head reached 0.0727 colour MSE against mean-copy’s 0.0696. More data narrowed nothing that mattered.

Both are symptoms of one mechanism. Completion is *ambiguous*: the occluded region is unobserved, so many colourings are consistent with the same visible evidence. A regressor trained under a squared-error loss is rewarded for predicting the *conditional mean* of that set of possibilities — so when it cannot tell red from blue, the loss actively pushes it toward washed-out grey. Mean-copy is not a weak opponent to be out-engineered; under an L2-type metric it is close to what the optimal *point* predictor does. You cannot train your way past it with a bigger regressor.

The principled escape is to stop asking for one error-minimising answer and instead learn a *distribution* over plausible answers, then draw one committed sample from it. That is the generative turn (ADR-0009), and this week is its first concrete instrument: a colour prior, adopted from our collaborator’s line of work (ADR-0010), tested honestly at scale.

2 The method

Geometry is not the problem here — a frozen, pretrained PoinTr already completes the shape, and we do not touch it. The open question is how the *filled-in points get coloured*.

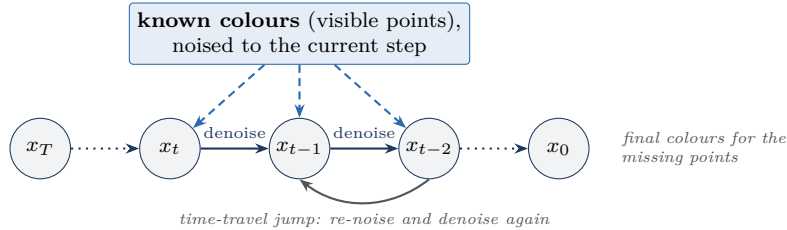


The central object is the **union cloud**: the visible partial points, whose colours we *know*, glued to PoinTr’s filled-in points, whose colours must be *invented*. Only the invented region is ever scored.

Background: diffusion, and how RePaint conditions it without retraining

A DDPM learns by destroying and rebuilding. Take real data, add a little Gaussian noise repeatedly ($T=200$ steps here) until nothing but noise remains; train a network to undo *one* step. To generate, start from pure noise and run those steps backwards. Here the diffused signal is **not geometry** — it is the per-point RGB field sitting on fixed, already-completed coordinates. The model learns “what do plausible aeroplane colourings look like?”

RePaint (Lugmayr et al., CVPR 2022) makes that unconditional model do inpainting *without ever training it on masks*. At each reverse step the known points are overwritten with the true colours, noised to that step’s level; the unknown points keep the model’s guess. Every so often the sampler jumps *back up* in noise and re-denoises (“time travel”), letting the invented region re-negotiate with the known one so the result agrees semantically, not just at the seam. One trained model then serves every occlusion ratio with no retraining — which is exactly why the occlusion sweep below is cheap.



“**Part-based**” is our collaborator’s addition, in three places: the denoiser pools features *within each part* (a learned analogue of the part-mean rule, making cross-part colour bleed structurally hard); a part with *zero* visible points is flagged *anchorless* and filled from the prior rather than from a neighbouring part’s colour; and the resampling budget grows with the worst part’s occlusion.

3 The experiment

3.1 Two stages, run as one controlled comparison

The same code, metric and protocol are run over two colour regimes, so that *prior-learnability* is the manipulated variable:

- **Synthetic palette** — each part is one flat colour plus a little noise (110 test models, 6,809 measurements). *Honest caveat, stated up front*: where every part is a single flat colour, part-mean with good labels is close to Bayes-optimal and intra-part texture gains cannot appear. This stage validates *the machine*, not the texture claim.
- **Real texture** — colours sampled from textured ShapeNet meshes (34 test models, 2,108 measurements). **This is the experiment.** The labelled data was built by us: 137 aeroplanes

The prize collapses on real texture: the whole part-colour model class has 8× less to win

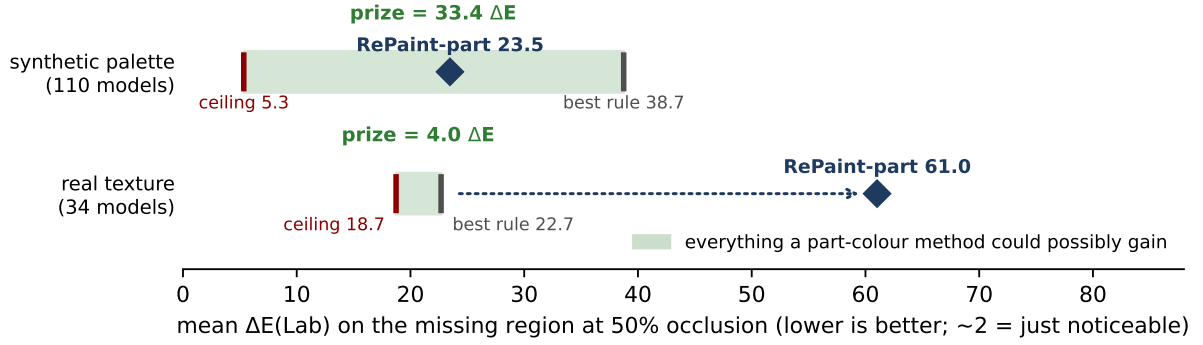


Figure 1: The prize collapses. The shaded band is everything a part-colour method could possibly gain: from the best trivial rule down to the oracle ceiling. On synthetic colours it spans 33.4 ΔE and RePaint-part lands inside it. On real texture it is only 4.0 ΔE wide — about two just-noticeable differences — and the method lands far outside it.

fused from our own HF bundles with ShapeNet-Part labels, frame-QA’d, refusing any model whose label transfer stays poor rather than writing bad labels.

3.2 Six rows, because a single number cannot be diagnosed

Every cell of every table reports the same metric — **mean $\Delta E(\text{Lab})$ over the missing region only**, on unseen test models. ΔE is perceptual colour distance; ≈ 2 is the threshold of visibility. Six rows are computed so that a result can be *explained*, not merely reported:

row	what it predicts	what it isolates
NN-copy	nearest visible point’s colour	trivial local copying
part-mean	mean visible colour of that part	the strongest trivial rule
oracle ceiling	part-mean, <i>ground-truth</i> labels	the best <i>any</i> part-colour method can do
RePaint-vanilla	the DDPM, no part conditioning	whether “part-based” earns its keep
RePaint-part	the method under test	—
RePaint-part (oracle seg)	the method, ground-truth labels	segmentation error

Two further axes: **E1** sweeps occlusion over {25, 50, 75}% — promised in the original notebooks but never run — and **E2** removes one part entirely (the *anchorless* case the method was designed for; the part is dropped by *ground-truth* label, because dropping by the segmenter silently degenerates when the segmenter never predicts that part). Because a sampler is structurally penalised by single-sample scoring, each diffusion row is drawn five times per model and reported as **mean-of-seeds** (the honest row), **best-of- k** (does the prior *cover* the truth?) and **diversity** (mean pairwise ΔE between samples).

4 Results

4.1 The headline: a collapsed margin, and a method outside it

Figure 1 is the week’s finding. On real texture the oracle ceiling sits at 18.7 while NN-copy already achieves 22.7: perfect part labels plus a perfect per-part average buy only 4.0 ΔE over naive copying. Real aeroplane texture is simply not well described by one colour per part — the residual is intra-part variation that no part-level model can represent. On synthetic colours the same measurement gives 33.4 ΔE of room, a 8.4× difference.

That single fact reframes the result. RePaint-part scoring 61.0 against 22.7 is not a near miss

inside a rich design space; it is a method competing for a margin that had already almost vanished, and overshooting it by an order of magnitude.

variant	synthetic palette		real texture	
	mean	best-of- k	mean	best-of- k
NN-copy	49.1	49.1	22.7	22.7
part-mean	38.7	38.7	24.5	24.5
oracle ceiling	5.3	5.3	18.7	18.7
RePaint-vanilla	39.8	37.3	62.2	55.6
RePaint-part	23.5	23.4	61.0	52.8
RePaint-part (oracle seg)	17.6	17.5	61.1	52.7

Table 1: Mean $\Delta E(\text{Lab})$ on the missing region at 50% occlusion (lower is better); best-of- k is the minimum over 5 seeds per model. Models won by RePaint-part against the better trivial rule: 99/110 synthetic, 0/34 real. Full sweep in Section A.

4.2 Occlusion: the behaviour inverts

Same code, opposite behaviour: what happens as more colour is hidden

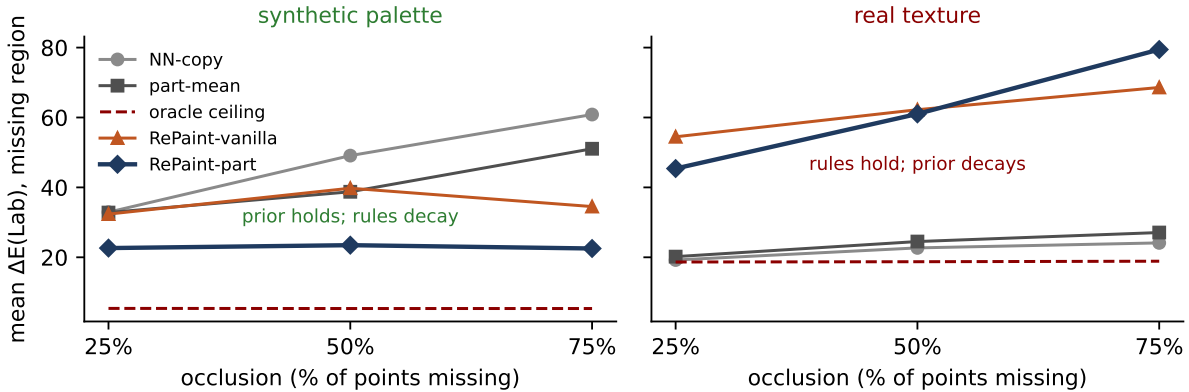


Figure 2: The occlusion sweep. As more colour is hidden, a genuine prior should matter *more* and copy rules should decay. That is exactly the synthetic picture (RePaint-part flat at 22.6→22.5 while NN-copy climbs 32.9→60.8). On real texture it inverts: the rules stay flat and the prior degrades 45.4→79.4.

This sweep is the most diagnostic axis we added, and it separates the two stages cleanly. When the prior is learnable, hiding more colour hurts the copy rules (less to copy from) and barely touches the prior — the classic argument for generative completion. When the prior is *not* learnable, hiding more colour simply hands more of the output over to a bad prior, so the method degrades while the rules, which still have local evidence, hold.

The anchorless case (**E2**, one part entirely missing) makes the same point at its extreme. It is the scenario the method exists for, and on synthetic colours it delivers crushingly: 16.8 against 57.5 for part-mean. On real texture it fails hardest of all: 84.6 against 17.8.

4.3 What it looks like

Figure 3 shows why the failure is interesting rather than embarrassing. The regression failures of s5/s6 looked like *washed-out* colour — hedging toward the mean. This failure looks the opposite: vivid and confident, and wrong. The model has learned that aeroplanes have liveries; it has not learned which livery *this* aeroplane has, because roughly a hundred training clouds cannot teach

One held-out test aeroplane per stage, 50% occluded (the two centre-right columns are the diffusion prior's answer)

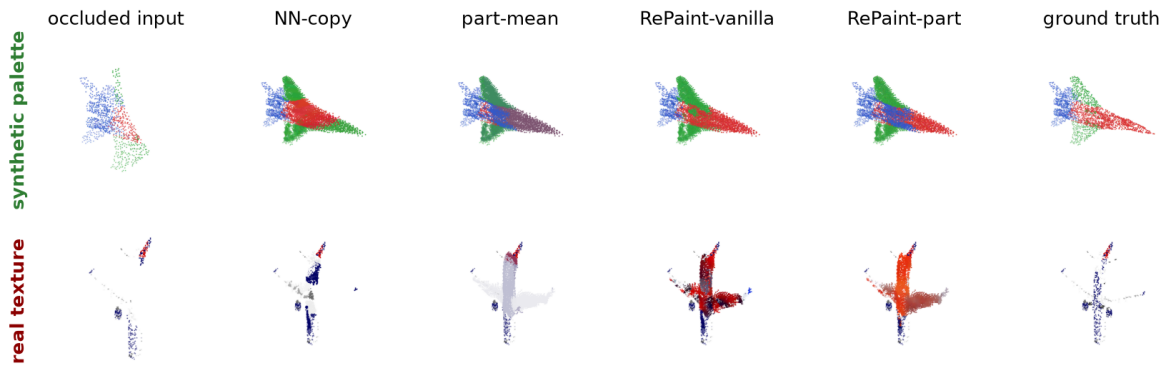


Figure 3: The mechanism, visible. Top row (synthetic): the prior reconstructs the part colours convincingly. Bottom row (real texture): the ground-truth aeroplane is predominantly white, and the prior confidently paints a red-orange livery over it. This is not blur or hedging — it is a committed, plausible, wrong answer, which is precisely what sampling from an under-trained prior produces.

that. Sampling replaced the hedging with commitment, exactly as intended — and commitment without knowledge is worse than hedging.

5 Why we trust the negative

A negative result is only worth reporting if the boring explanations are excluded. Four controls, each built into the protocol in advance, do that:

1. **Not the segmenter.** Given *ground-truth* part labels the method scores 61.1 — statistically indistinguishable from 61.0 with predicted labels. This matters because it is precisely the confound that made the collaborator’s first pass inconclusive (a segmenter that never predicted one of the parts); removing it leaves the result unchanged.
2. **Not sampling luck.** Taking the *best* of five independent draws per model still gives 52.8, more than twice the best trivial rule. The prior does not merely miss on average — it does not cover the right answer at all.
3. **Not a near miss.** Sample diversity is $31.4\Delta E$ on real texture against 9.2 on synthetic. Since ≈ 2 is just-noticeable, the real-texture samples are scattered across wildly different colour schemes: an uncommitted prior, not a tight distribution centred slightly off-target.
4. **Not the code.** The identical pipeline wins 99/110 models on synthetic colours. Training, sampling, conditioning and metric all work.

The mechanism that survives is the simple one: **an unconditional colour prior trained on of the order of a hundred real-texture clouds is too weak, and sampling-time conditioning cannot extract structure the prior never learned.** RePaint can re-inject known colours and harmonise around them; it cannot invent knowledge of *this* object’s colour scheme.

6 What the scaling bought us

Running this as a formal experiment rather than a demonstration produced three findings that a 4-model, single-seed pass could not have shown.

- **The headroom collapse** (Section 4.1) — the structural result, and the one that generalises beyond this method: on real ShapeNet texture, *any* model that assigns one colour per part is

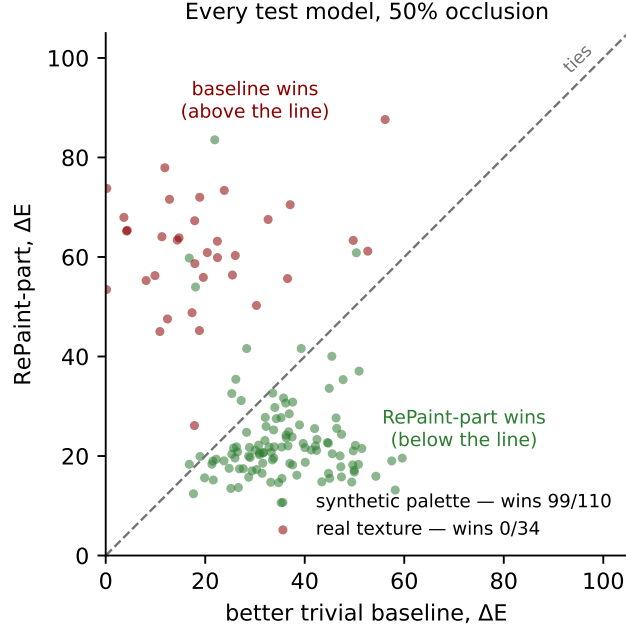


Figure 4: Every test model. Points below the diagonal are models where RePaint-part beat the better trivial rule. The two stages separate almost completely; the real-texture loss is not an average dragged down by a few outliers.

confined to a 4.0 ΔE margin.

- **Part-conditioning flips sign** (Figure 6). On synthetic colours it helps substantially at every occlusion level of E1 (23.5 vs. 39.8 at 50%), but its benefit decays as occlusion grows, and in the anchorless case it is already *negative* (16.8 vs. 11.2). On real texture the reversal comes earlier and harder: helpful at 25%, neutral at 50%, clearly harmful by 75% (79.4 vs. 68.6) and worst at E2 (84.6 vs. 65.4). The reading is that part-pooling makes the model *commit harder* to a part-level colour decision — which is a gain when the part colour is learnable and a penalty when it is not.
- **Diversity is measurable, and it is the tell.** The $\approx 3.4\times$ difference in sample spread between regimes (9.2 vs. 31.4) is the most direct read-out of prior quality we have, and it comes free from the same samples.

7 Honest read

- **What we can claim.** Within one category (aeroplanes), at this data scale, on real texture, part-based RePaint colour inpainting does not beat trivial local colour rules, and the failure is attributable to prior weakness rather than to segmentation, sampling variance, or implementation. Independently: the part-colour model class itself has 4.0 ΔE of headroom on this data.
- **What we cannot claim.** That the method is bad in general. It is untested at larger data scales, on other categories, and with trained (rather than sampling-time) conditioning. Our real-texture set is 137 models of one category; the prior is trained on roughly a hundred of them. Nothing here says the approach fails with $10\times$ the data — only that it fails with this much.
- **What the synthetic stage does and does not prove.** It proves the machine is correct and that RePaint’s occlusion behaviour is real. It cannot prove a texture claim, because part-mean is near-optimal there by construction.
- **On the collaborator’s earlier numbers.** Her real-texture first pass (4 models, 1 seed, with

Four controls, four mundane explanations ruled out

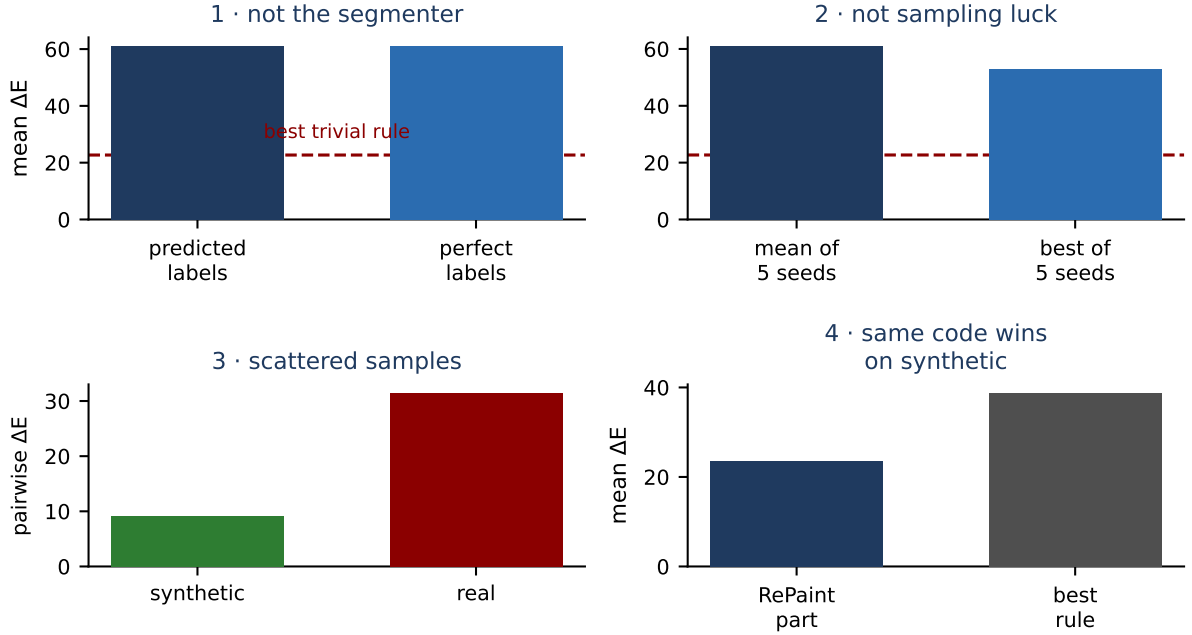


Figure 5: Four controls. Each rules out one ordinary explanation for the loss.

a part-segmenter that never predicted one of the parts) was explicitly not presented as a result, and this work does not contradict it — it supplies the verdict that setup could not.

phase	what predicted colour	its trivial baseline	beat it?
s5	decoupled learned colour head	mean-copy	no
s6	the same head, $N=150/\text{cat}$	mean-copy (0.0696 vs. 0.0727)	no ($\times 1.045$)
s7	RePaint diffusion colour prior	NN-copy (22.7 vs. 61.0)	no ($\times 2.69$)

Table 2: The through-line. Three different mechanisms — a regression head, the same head with more data, and a generative prior — have now failed to beat trivial local colour statistics on real texture. *Caveat:* s6 is RGB-MSE and s7 is $\Delta E(\text{Lab})$, measured in different pipelines; the ratios are within-phase and dimensionless, and the pattern is qualitative. Cross-pipeline magnitudes are not comparable (ADR-0007). s5’s magnitude is superseded and is quoted only as an outcome.

Table 2 is the most useful thing to carry forward. The consistent obstacle is not any one architecture; it is that per-point colour on real ShapeNet texture appears to be dominated by object-specific information that is not learnable, as a transferable rule, at the scales we can currently reach. Two of the three attempts failed by hedging; the third failed by committing. That is a statement about the data, and it is worth more than another architecture would have been.

8 Proposed next steps

- 1. More textured training data — the top lever.** Everything above points here. 137 aeroplanes support a prior trained on ~ 100 clouds; 107 chairs and 51 cars are already label-capable through the same fusion path, and the background data generator can extend the bundles. This is the one change the diagnosis directly recommends.
- 2. Trained conditioning instead of sampling-time only.** Show the mask during training.

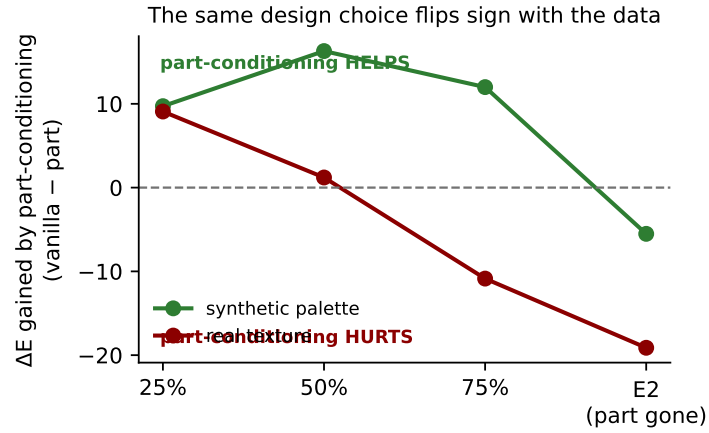


Figure 6: Part-conditioning is not universally good. Its advantage shrinks as occlusion grows in *both* regimes, and on real texture it crosses into active harm.

This is also the bridge to ADR-0009’s geometry-diffusion spine, and it gives a second conditioning mechanism to compare against RePaint on identical data.

3. **Condition on a per-object colour scheme.** The panel failure is a model that knows aeroplanes have liveries but not which one; a global colour code would remove exactly that ambiguity, and it is cheap to test.
4. **Escape the part-colour class.** Given a 4.0 ΔE ceiling, any further work confined to one-colour-per-part is capped. Intra-part texture modelling is where the remaining error actually lives.
5. **Deprioritised: segmentation repair**, capacity and T tuning — the oracle-seg control already shows these are not the binding constraint.

How this week was run

Roles. Method and original notebooks: Pelin Kılıç. Efe: direction, decisions at forks, and the Colab runs. Claude (AI co-researcher): the production port and its tests, the fused labelled dataset, experiment and control design, analysis, and this report. **The loop** (`docs/WORKFLOW.md`): decide → grill the plan → ADR → build + test locally → run resiliently on Colab → analyse → document + report. **This fortnight in numbers:** 1 ADR (0010), 2 tested modules (18 unit tests, all green before any GPU time), 1 notebook (18 cells, both stages), 137 labelled models fused by us, 6,809 + 2,108 evaluation measurements, zero lost work across VM recycles. Timeline and incidents: Section D.

A Full sweep — both stages, all occlusion levels

table	variant	synthetic palette			real texture		
		25%	50%	75%	25%	50%	75%
E1	NN-copy	32.9	49.1	60.8	19.2	22.7	24.1
E1	part-mean	32.8	38.7	51.1	20.2	24.5	27.1
E1	oracle ceiling	5.4	5.3	5.3	18.7	18.7	18.9
E1	RePaint-vanilla	32.4	39.8	34.5	54.5	62.2	68.6
E1	RePaint-part	22.6	23.5	22.5	45.4	61.0	79.4
E1	RePaint-part (oracle seg)	16.1	17.6	15.8	44.4	61.1	77.4
E2	NN-copy	–	65.8	–	–	25.3	–
E2	part-mean	–	57.5	–	–	17.8	–
E2	RePaint-vanilla	–	11.2	–	–	65.4	–
E2	RePaint-part	–	16.8	–	–	84.6	–
E2	RePaint-part (oracle seg)	–	18.3	–	–	86.4	–
DIV	RePaint-vanilla	18.1	18.2	18.2	46.7	50.5	53.5
DIV	RePaint-part	9.7	9.2	8.4	26.5	31.4	25.4
DIV	RePaint-part (oracle seg)	9.6	9.2	8.2	27.5	32.3	25.6

Mean $\Delta E(\text{Lab})$ over the missing region, unseen test models only. **E1** = occlusion sweep; **E2** = one part fully occluded (50 % crop, dropped by ground-truth label; 109 of 110 synthetic models have a droppable part); **DIV** = mean pairwise ΔE between the 5 samples, a diversity measure for which neither “high” nor “low” is inherently good — it reads as commitment when the prior is accurate and as scatter when it is not. The oracle ceiling has no E2 row: with a part entirely unobserved, a part-mean oracle has no anchor to average.

B Who contributed what

element	origin
Part-based RePaint colour inpainting; original notebooks	Pelin Kılıç
RePaint sampling-time conditioning	Lugmayr et al., CVPR 2022
DDPM formulation	Ho et al., NeurIPS 2020
Geometry backbone (frozen, pretrained)	PoinTr; her fork, pinned by us
Production port + unit tests (<code>tools/pc_repaint.py</code>)	this project
Labelled real-texture dataset (<code>tools/pc_partlabels.py</code>)	this project
Control design: oracle-seg, best-of- k , diversity, occlusion sweep	this project
Scaled evaluation, verdict, and this report	this project

The method is hers and is reproduced faithfully — same architecture, same ΔE protocol, same six rows. What this project added is scale (34 test models $\times$ 5 seeds rather than 4×1), the controls that make a negative interpretable, a labelled dataset that depends on no private storage, and the evaluation discipline. The HF dataset is public, so every artifact here is retrievable without our credentials.

C Provenance & how to reproduce

- **Code:** <https://github.com/efeyenice/point-cloud-completion-research> — `tools/pc_repaint.py` (DDPM, RePaint sampler, part-conditional denoiser, ΔE metrics, config-signed checkpoints, resumable eval rows; 13 tests), `tools/pc_partlabels.py` (ShapeNet-Part label fusion with 48-frame QA; 5 tests), `notebooks/repaintTraining.ipynb` (both stages via `DATA_SOURCE`). Decision record: `docs/adr/0010-adopt-repaint-color-line.md`.
- **Artifacts:** HF dataset [efeyenice/pc-completion-data](#) at revision 2fa9f9b821 — `runs/repaint/eval_rows_auto.csv` (6,809 rows), `runs/repaint/eval_rows_labeled_s3.csv`

(2,108 rows), the two qualitative panels, config-signed checkpoints under `runs/repaint/ckpts/`, the 137 fused labelled models under `data_labeled/02691156/`, and the mirrored PoinTr weights.

- **Protocol:** single category (aeroplane, synset 02691156); colour clouds at 2048 points; 25 % of models held out for test, never trained on; DDPM $T=200$ steps, cosine schedule, width 128, $k=16$, 3 blocks, 300 epochs; 5 sampling seeds per model; occlusion by PoinTr’s viewpoint crop.
- **Reproduce:** open `notebooks/repaintTraining.ipynb` on Colab (GPU), set `DATA_SOURCE`, add `HF_TOKEN + WANDB_API_KEY` to Secrets, run all. Evaluation is resumable per row, so re-running is always safe. This report: `make figures && make in reports/2026-08-19-repaint-color/` — every number in the prose is regenerated from the two CSVs above.

D Process log

Timeline

- **Aug 7.** Advisor endorsed both the scale-curve report and the collaborator’s RePaint notebooks. Mandate: replicate at scale under production workflow. Grilled to a plan over three rounds; ADR-0010 written; both stages of the port built and unit-tested on CPU before any GPU time.
- **Aug 7–12.** Self-sufficiency work: because the collaborator’s labelled dataset and build scripts exist only in her personal storage, the labelled real-texture set was rebuilt from our own bundles plus the ShapeNet-Part benchmark. The fusion driver refuses models whose label transfer fails QA rather than writing them silently.
- **Aug 12–13.** Both stages run on Colab. Synthetic gate passed first and reproduced the expected ordering, at roughly $11\times$ the original evaluation scale; the real-texture stage then ran to completion. Every measurement is one CSV row synced to HF, so interruptions cost at most the row in flight.
- **Aug 13–19.** Results verified by recomputing both stages’ tables from the raw CSVs; the headroom collapse was found during that verification. This report written from the artifacts, with all figures and numbers scripted.

Incidents → lessons

1. **An import collision.** PoinTr ships its own regular `tools` package, which shadows ours regardless of `sys.path` order. Fixed by importing our modules `flat`; recorded so it is not re-broken.
2. **Checkpoints are dangerous when silent.** A checkpoint trained on synthetic colours will load without complaint into a real-texture run and quietly invalidate the whole table. Every checkpoint is therefore signed with its full config and rejected on mismatch — an idea taken directly from the collaborator’s notebooks and kept.
3. **Ablations must drop by ground truth.** Removing a part using the segmenter’s labels does nothing when the segmenter never predicts that part, so the experiment silently becomes a no-op. E2 drops by ground-truth label for exactly this reason.
4. **Validate the machine on data where the answer is known.** The synthetic stage cost little and is what makes the real-texture negative interpretable rather than ambiguous.